

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458770.847 +/- 0.019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.049
Transit depth (Rp/Rs)^2	1.53 +/- 1.22 [%]
Orbital Inclination (inc)	89.67 +/- 2.86
Airmass coefficient 1 (a1)	7.11 +/- 2.3
Airmass coefficient 2 (a2)	-1.53 +/- 1.67
Scatter in the residuals of the lightcurve fit is	29.89 %
Best Comparison Star	#3 - [460, 364]
Optimal Aperture	3.2
Optimal Annulus	7.68
Transit Duration (day)	0.036 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.124 ± 0.049 vs 0.1404 (deviation 0.33σ)
Duration fit vs published	0.036 d vs 0.0754 d (deviation 2.32σ)
Mid-time O–C	24.1 min
Depth SNR	2.5
Residual scatter	29.89 %

Light curve

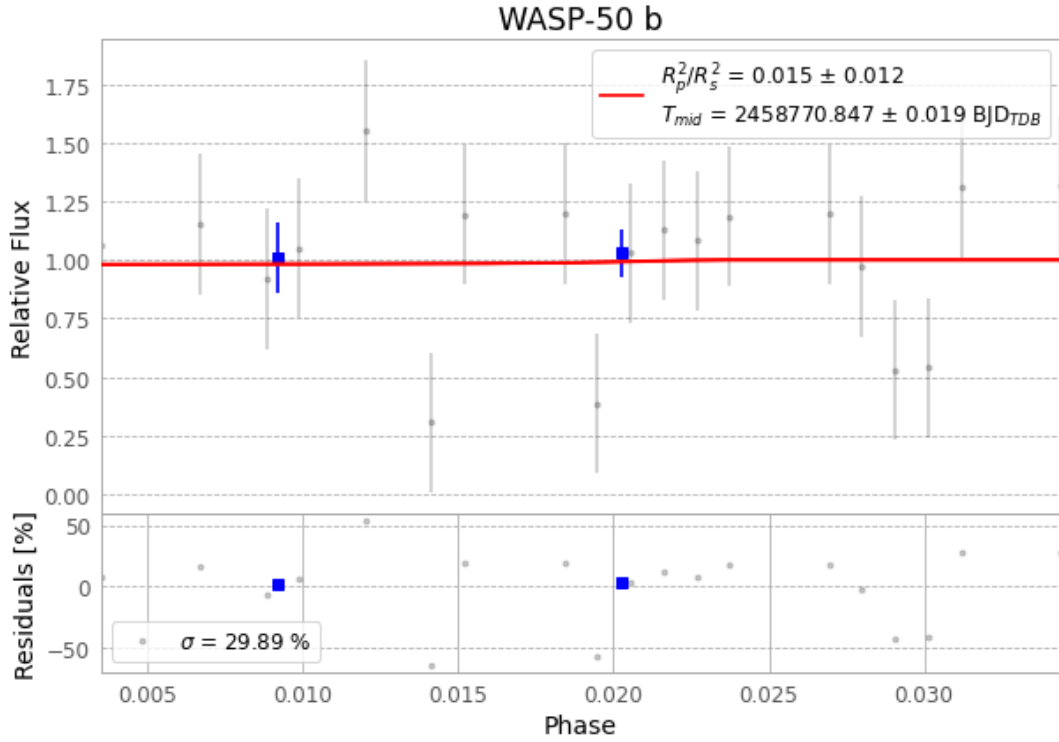


Figure 1 — FinalLightCurve_WASP-50 b_2019-10-14.png (SHA-256 0c009fbf8d3a0609...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [423, 114], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fab3954c2b9b6949...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

32 FITS frames (21 MB), WASP-50191014081815.FITS → WASP-50191014095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-191014.log (2c3c28b72a08...), FinalLightCurve_WASP-50 b_2019-10-14.png (0c009fbf8d3a...), FinalLightCurve_WASP-50 b_2019-10-14.csv (12e33843457f...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 16:27Z.